RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)
Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

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Branch: B.E. Computer Science and Engineering (Cyber Security)

Year / Semester: III Year / V Semester

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Academic Year: 2026 - 2027

INDEX

Experiment No.	Title of the Experiment	Date
1	Study of Computer Forensics and Different Tools Used for Forensic Investigation	21-07-2026
2	Introduction to Digital Forensics (TryHackMe)	28-07-2026
3	Windows Forensics 1 (TryHackMe)	28-07-2026
4	Live Forensics Case Investigation using Autopsy	04-08-2026
5	Recover Deleted Files using Forensics Tools	11-08-2026
6	Find Last Connected USB on the System (USB Forensics)	18-08-2026
7	View Last Activity of Your PC	25-08-2026
8	Extraction of Browser Artifacts	01-09-2026
9	Comparison of Two Files by Compare It Software	08-09-2026
10	Collecting Email Evidence in Victim PC	15-09-2026
11	Hiding and Extracting a Text File behind an Image / Audio File using Command Prompt	29-09-2026
12	Extraction of EXIF Data from Image Files using ExifReader	01-10-2026

EXPERIMENT – 1

Study of Computer Forensics and Different Tools Used for Forensic Investigation

Aim

To study the field of computer (digital) forensics and to understand the tools used at the different stages of a forensic investigation.

Introduction

Digital forensics is the discipline of identifying, preserving, analysing and presenting digital evidence in order to establish who was responsible for a digital intrusion or computer crime. It rests on Locard's exchange principle: whenever a crime is committed, the offender leaves traces behind. In a digital environment these traces include log entries, registry changes, remnants of deleted files, malware and attacker tools, all of which help to reconstruct events and attribute them to an individual.

Investigators generally use three major commercial suites – EnCase Forensic (Guidance Software), Forensic Toolkit (FTK, AccessData) and ProDiscover. These are expensive and mostly used by law enforcement. Comparable capabilities are available in free and open-source tools such as The Sleuth Kit, Autopsy, Helix and Knoppix, and in distributions such as BackTrack and Kali Linux, although with a less polished interface.

Procedure

- Identified the commercial forensic suites used by law enforcement: EnCase Forensic, FTK and ProDiscover.
- Studied the open-source suites: The Sleuth Kit, Autopsy, Helix and Knoppix.
- Listed the individual tools bundled with Kali / BackTrack and noted the purpose of each.
- Studied the field tools FTK Imager, Log Parser Lizard GUI, Noxcivis Field Toolkit, Active@ Partition Recovery and Autopsy.
- Grouped the tools by investigation stage, as summarised in Table 1.1.

Table 1.1 Forensic tools studied and their purpose

Category	Tool	Purpose
Commercial suite	EnCase Forensic	Acquisition, analysis and reporting; widely used by law enforcement
Commercial suite	FTK (Forensic Toolkit)	Indexed evidence processing and analysis
Commercial suite	ProDiscover	Disk imaging and forensic investigation
Open source	The Sleuth Kit (TSK)	Command-line tools for file-system and disk-image analysis

Category	Tool	Purpose
Open source	Autopsy	Graphical front-end for TSK; case management and reporting
Open source	Helix / Knoppix	Bootable live forensic environments
Acquisition	FTK Imager	Forensic disk imaging, memory capture and file recovery
Acquisition	DumpIt	Physical memory dump of Windows systems
Analysis	Bulk Extractor	Extracts e-mail, URLs and phone numbers from images
Analysis	Log Parser Lizard GUI	Query and analysis of log files
Live response	Noxcivis Field Toolkit	Collects volatile data from a live computer
Recovery	Active@ Partition Recovery	Recovers deleted and lost partitions
Recovery	Recover My Files	Recovers deleted files from drives and images
Artifacts	USBDeview	Lists connected and previously used USB devices
Artifacts	LastActivityView	Timeline of recent user and system actions
Artifacts	ChromeHistoryView / IECacheView	Browser history and cache analysis
Utility	Compare It	Side-by-side file comparison
Metadata	ExifReader	EXIF metadata extraction from images

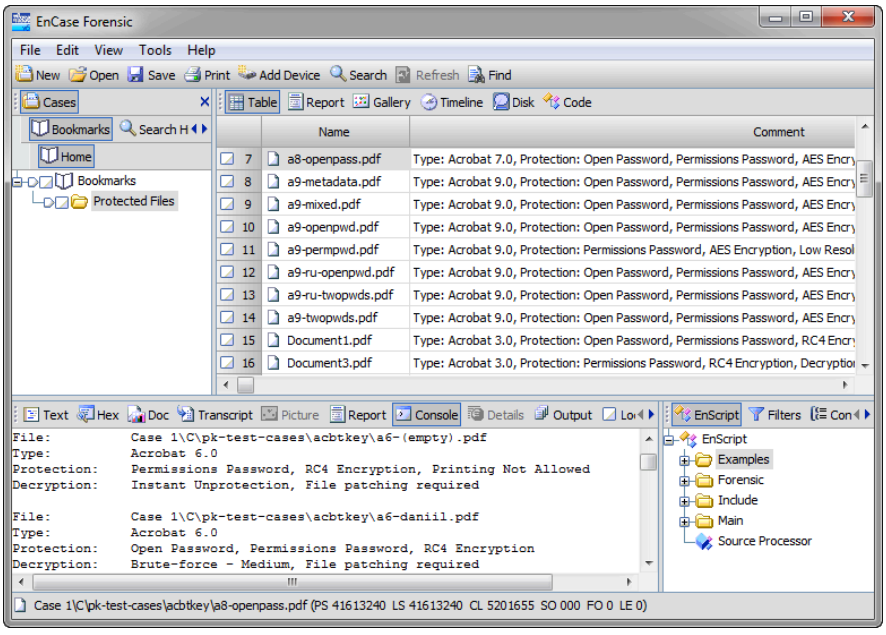


Fig. 1.1 EnCase Forensic – commercial forensic suite (reference illustration)

Time Stamp: 11-10-2026, 09:07:54

Table 1.2 Selected forensic tools available in the BackTrack / Kali distribution

Kali / BackTrack tool	Use
sleuthkit, autopsy	Disk and file-system analysis
scalpel, foremost-style carvers	File carving from raw data
dc3dd	Forensic disk imaging with hashing
exiftool	Metadata extraction
hexedit	Hexadecimal inspection of files
rifiuti2	Recycle Bin artifact parsing
evtparse.pl	Windows event log parsing
fatback	FAT file recovery
driftnet, ptk, iphoneanalyzer, timestomp, truecrypt	Network image capture, GUI analysis, mobile backups, timestamp analysis and encrypted volumes

Result

Thus, the field of computer forensics was studied and the commercial, open-source and distribution-bundled tools used for acquisition, analysis, recovery and reporting were identified along with their purposes.

EXPERIMENT – 2

Introduction to Digital Forensics (TryHackMe)

Aim

To understand the fundamentals of digital forensics and the standard forensic process by completing the TryHackMe room "Intro to Digital Forensics".

Procedure

- **Task 1 – Introduction:** Studied the definition of digital forensics and where it is applied (criminal investigation, corporate incident response and civil cases).
- **Task 2 – Digital Forensics Process:** Studied the standard phases – identification, preservation, collection, examination, analysis and reporting.
- **Task 3 – Practical Example:** Analysed a sample scenario to trace the source of the evidence.
- Answered the questions in each task and confirmed that the room showed 100% completion.

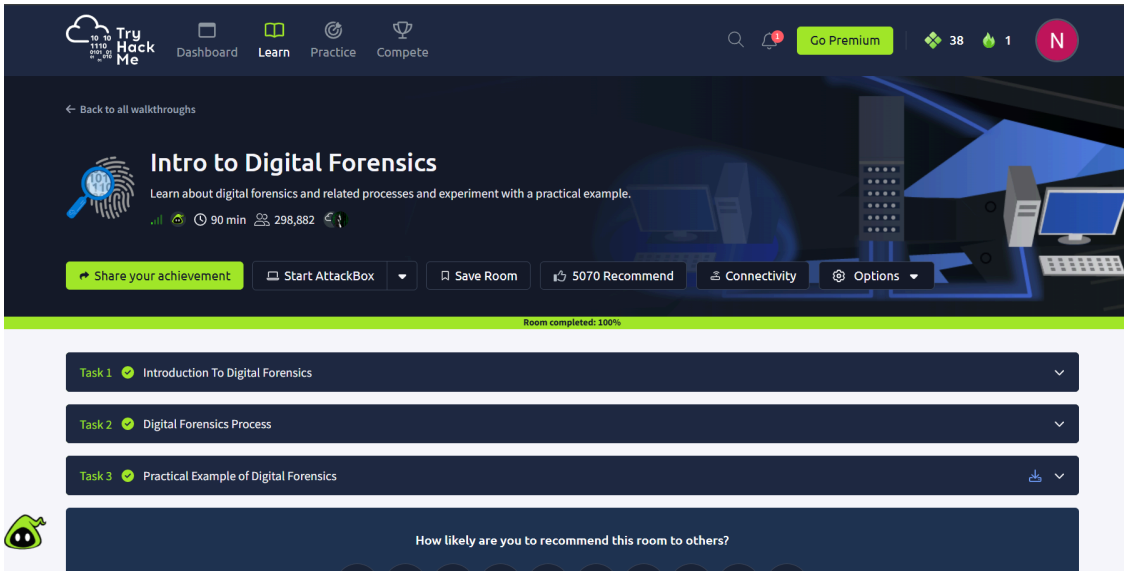


Fig. 2.1 TryHackMe – "Intro to Digital Forensics" room at 100% completion
Timestamp: 11-10-2026, 09:24:12

Result

The "Intro to Digital Forensics" room was completed with 100% progress. The basic concepts of digital forensics and the standard forensic process were understood.

EXPERIMENT – 3

Windows Forensics 1 (TryHackMe)

Aim

To learn Windows registry forensics and identify key Windows forensic artifacts by completing the TryHackMe room "Windows Forensics 1".

Procedure

- **Introduction and Registry:** Studied why Windows is a rich evidence source and the registry hives – SAM, SYSTEM, SOFTWARE, SECURITY, NTUSER.DAT and USRCLASS.DAT.
- **Offline access and acquisition:** Learned how hives are acquired from live and dead systems and examined offline with Registry Explorer.
- **System information and accounts:** Extracted hostname, last shutdown time, user accounts and logon details.
- **File and folder usage:** Studied recent files, shellbags and jump lists.
- **Evidence of execution:** Studied Run keys, prefetch files and related artifacts.
- **USB forensics:** Studied the registry keys that record external device connections.
- Completed the hands-on challenge and the conclusion.

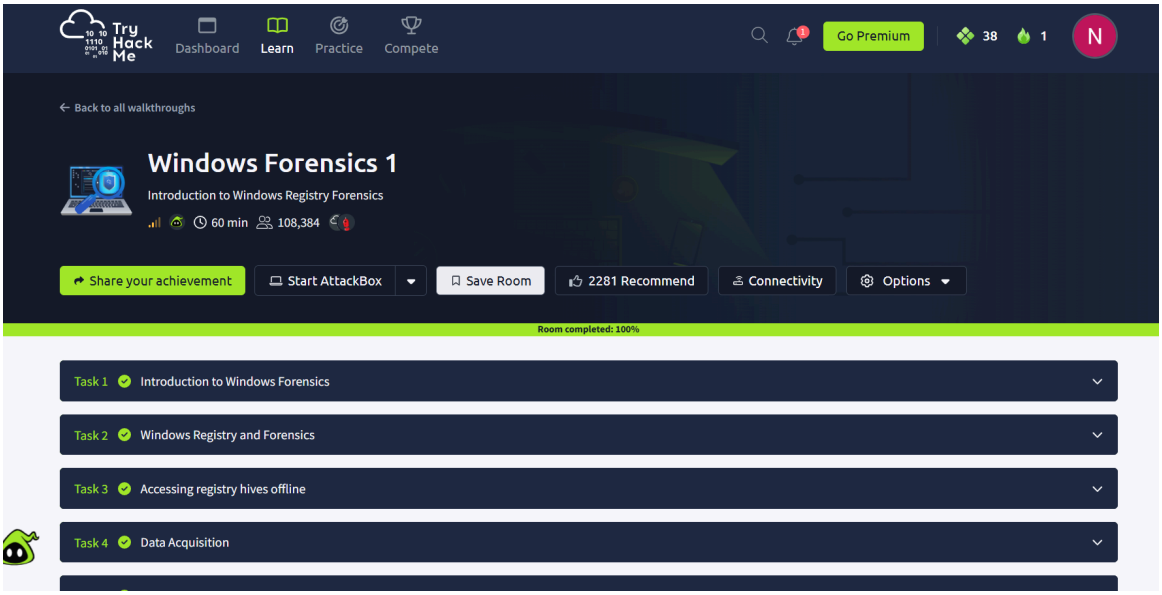


Fig. 3.1 "Windows Forensics 1" room – task list (Tasks 1 to 7)
Timestamp: 14-10-2026, 21:06:36

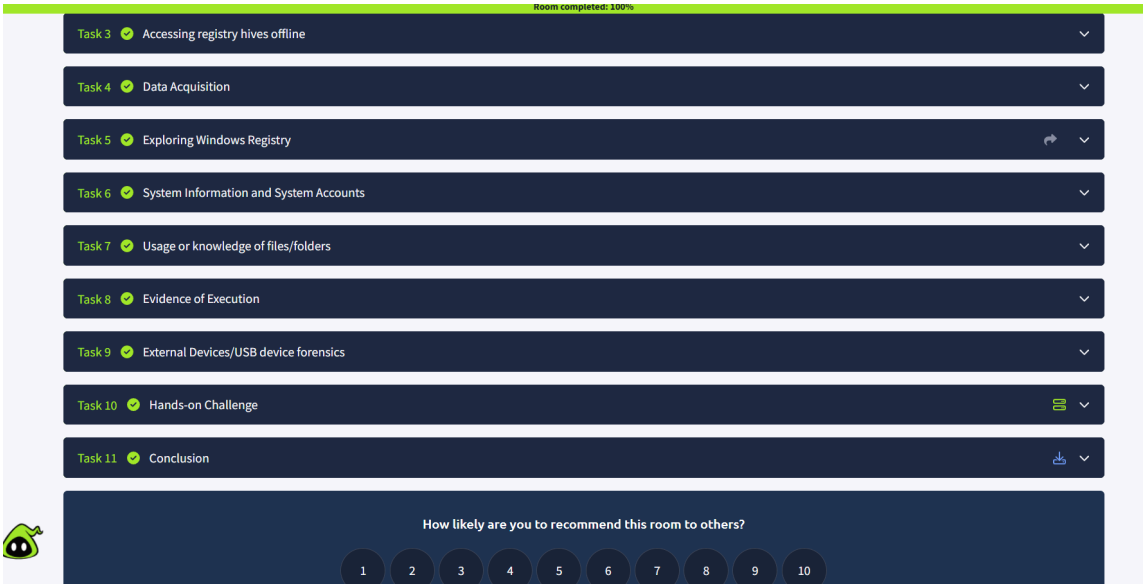


Fig. 3.2 "Windows Forensics 1" room – Tasks 8 to 11 including the hands-on challenge
Timestamp: 14-10-2026, 21:06:50

Result

The "Windows Forensics 1" room was completed at 100%. Registry hives, evidence of execution, file and folder usage artifacts and USB device artifacts were studied.

EXPERIMENT – 4

Live Forensics Case Investigation using Autopsy

Aim

To create a forensic case in Autopsy, add data sources, run the ingest modules and generate a forensic report.

Procedure

- Installed Autopsy 4.23.1 and launched it; selected ‘New Case’.
- Entered the case name and base directory, then the case number and examiner details (Fig. 4.1).
- In the Add Data Source wizard, selected the host and the data source type (Fig. 4.6).
- Browsed to the source, set the input time zone to Asia/Calcutta and the image split option (Fig. 4.7).
- Enabled all ingest modules – Hash Lookup, File Type Identification, Extension Mismatch Detector, Embedded File Extractor, Picture Analyzer, Email Parser, Encryption Detection, Interesting Files Identifier, PhotoRec Carver and Virtual Machine Extractor.
- Opened ‘Generate Report’, selected the HTML Report module and set the header and footer (Fig. 4.2).
- Reviewed the generated report: case summary, data sources, software versions and ingest history (Figs. 4.3 to 4.5).

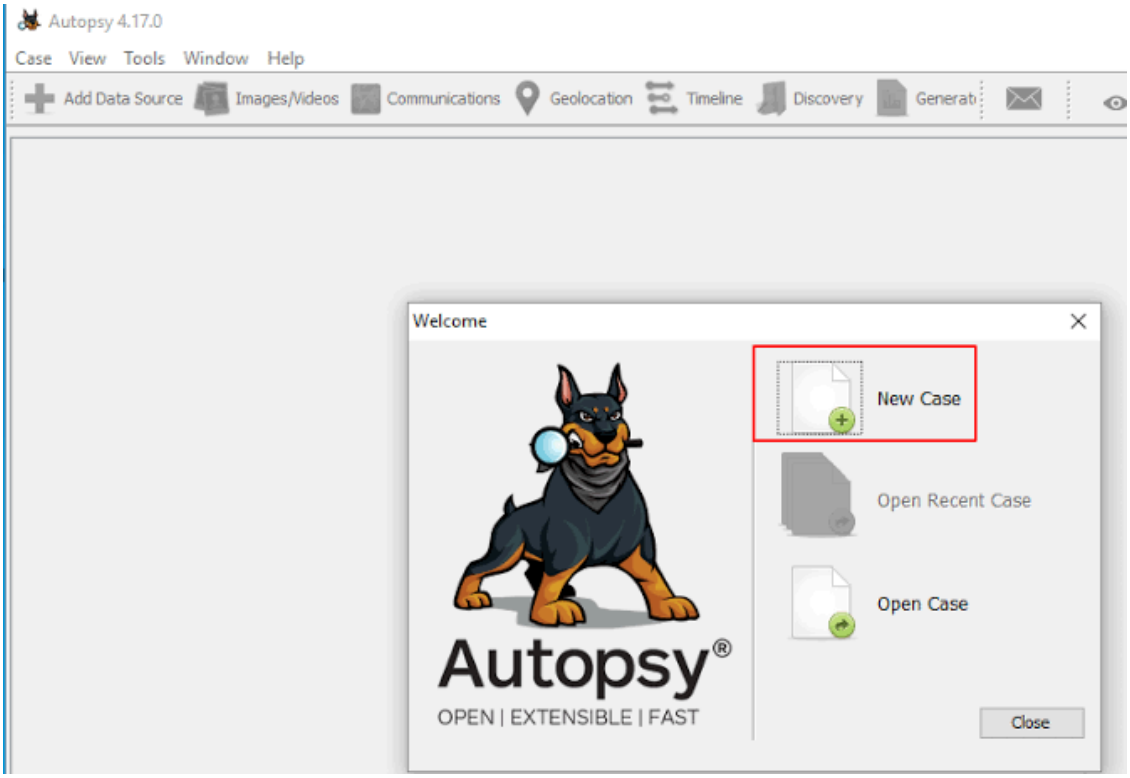


Fig. 4.1 Autopsy – case number and examiner details
Timestamp: 11-10-2026, 10:10:09

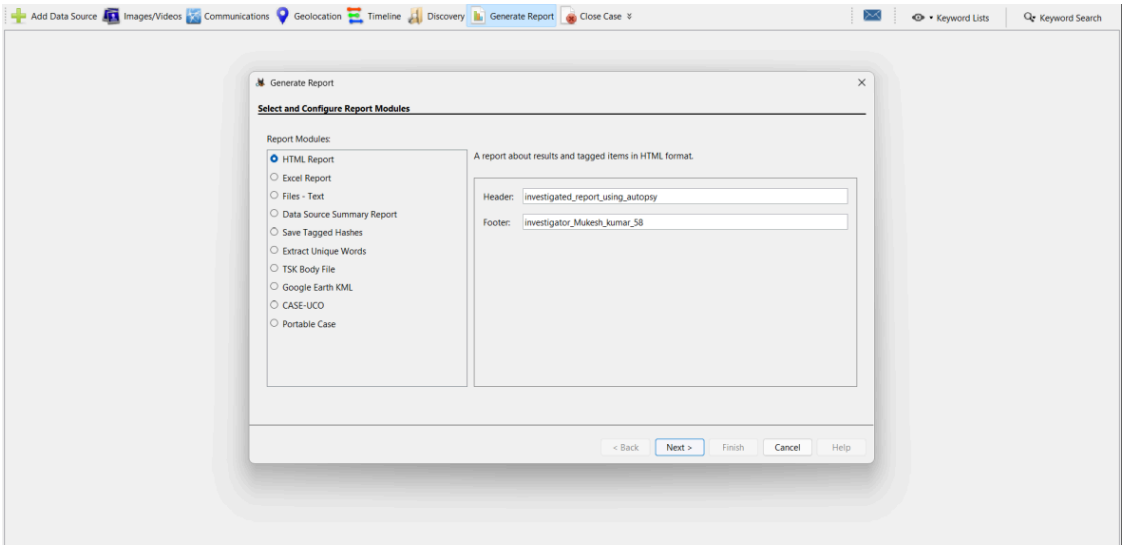


Fig. 4.2 Autopsy – selecting the HTML report module
Timestamp: 11-10-2026, 10:11:47

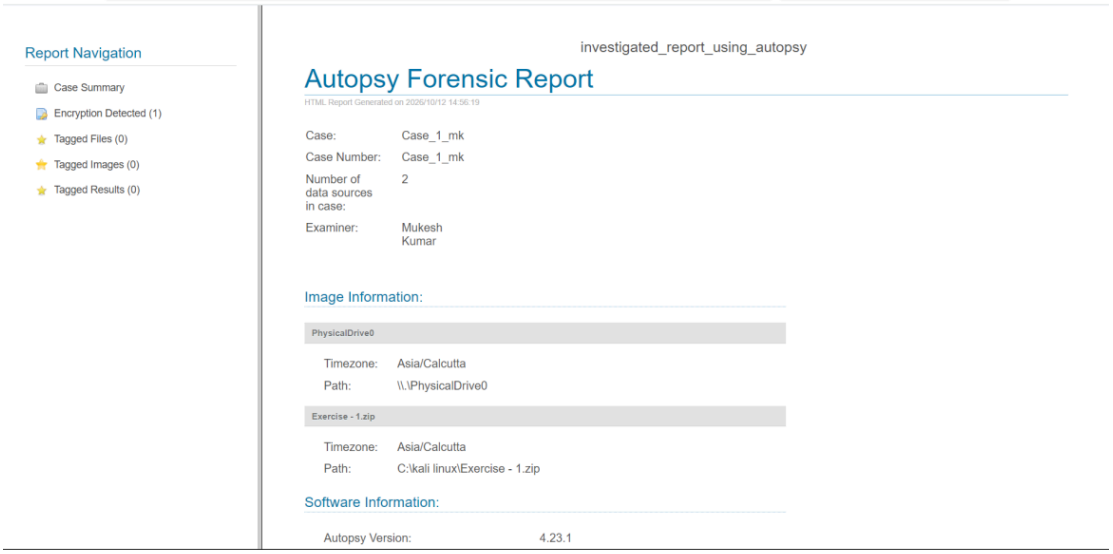


Fig. 4.3 Autopsy report – case summary and data sources (PhysicalDrive0 and Exercise - 1.zip) Timestamp: 11-10-2026, 10:15:35

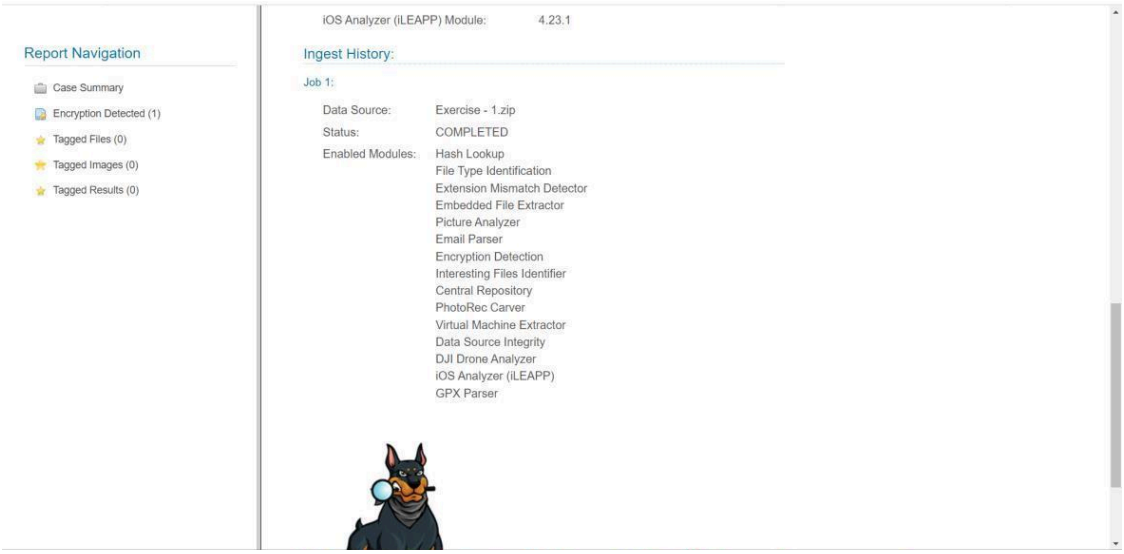


Fig. 4.4 Autopsy report – ingest history with status COMPLETED and enabled modules Timestamp: 11-10-2026, 10:17:02

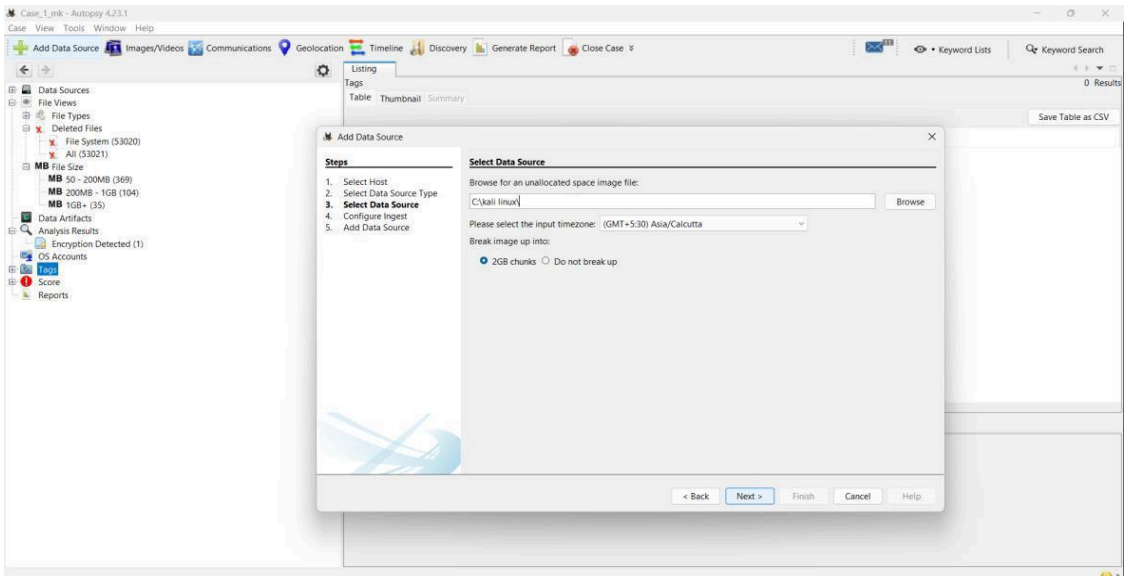


Fig. 4.5 Autopsy – selecting the image file and input time zone
Timestamp: 11-10-2026, 10:22:10

Result

Thus, an Autopsy case was created with two data sources, the ingest modules were run to completion and an HTML forensic report was generated. The report recorded the case summary, data sources, software versions and ingest history, and flagged one item under Encryption Detected.

EXPERIMENT – 5

Recover Deleted Files using Forensics Tools (FTK Imager)

Aim

To create a forensically sound image of a drive using FTK Imager and verify its integrity using hash values, so that deleted data can be examined and recovered.

Procedure

- **Create the file:** Created ‘malicious.txt’ in Notepad with sample text and saved it .
- **Delete the file:** Deleted the file. Its data remains in unallocated space until overwritten.
- **Why image first:** Deleted files are not captured by normal copy or backup, so a bit-by-bit image is acquired first and examined instead of the original.
- **Start imaging:** Opened FTK Imager 3.1.2.0 and selected File → Create Disk Image.
- **Source:** Selected Physical Drive as the evidence type .
- **Image type:** Chose the destination image type – Raw (dd), SMART, E01 or AFF.
- **Evidence details:** Entered the case number, evidence number, description, examiner and notes.
- **Destination:** Set the destination folder, image file name, fragment size and compression
- **Acquire:** Clicked Start and monitored the progress.
- **Verify:** Used File → Verify Drive/Image to compare the computed and stored MD5 and SHA-1 hashes.
- **Recover:** Add the image as an evidence item, locate the deleted file (marked with a red cross), and export it to compare with the original content.

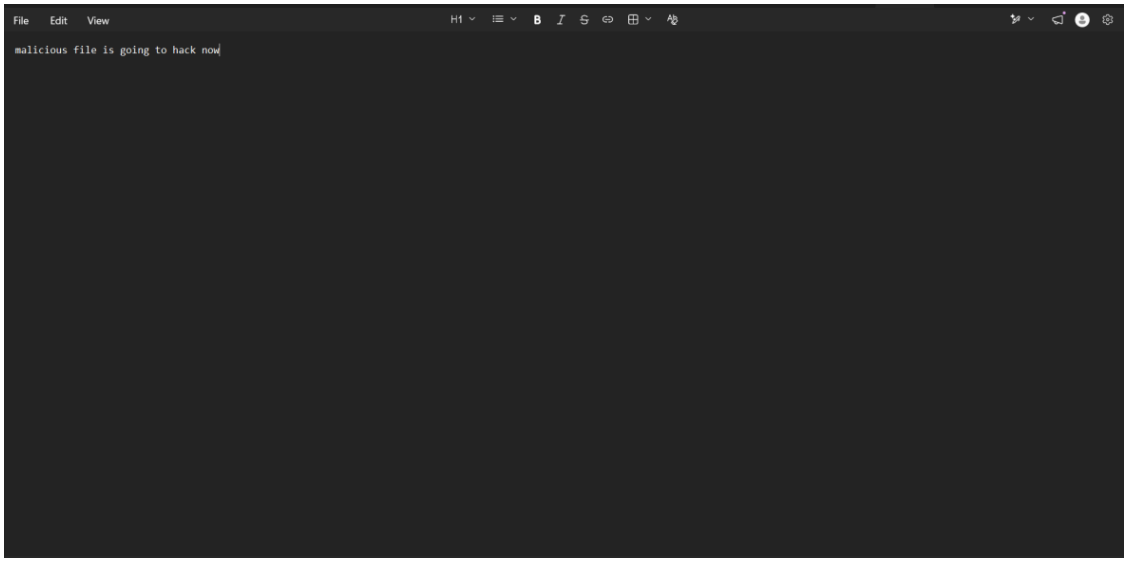


Fig. 5.1 Creating the file 'malicious.txt'
Timestamp: 11-10-2026, 10:30:12

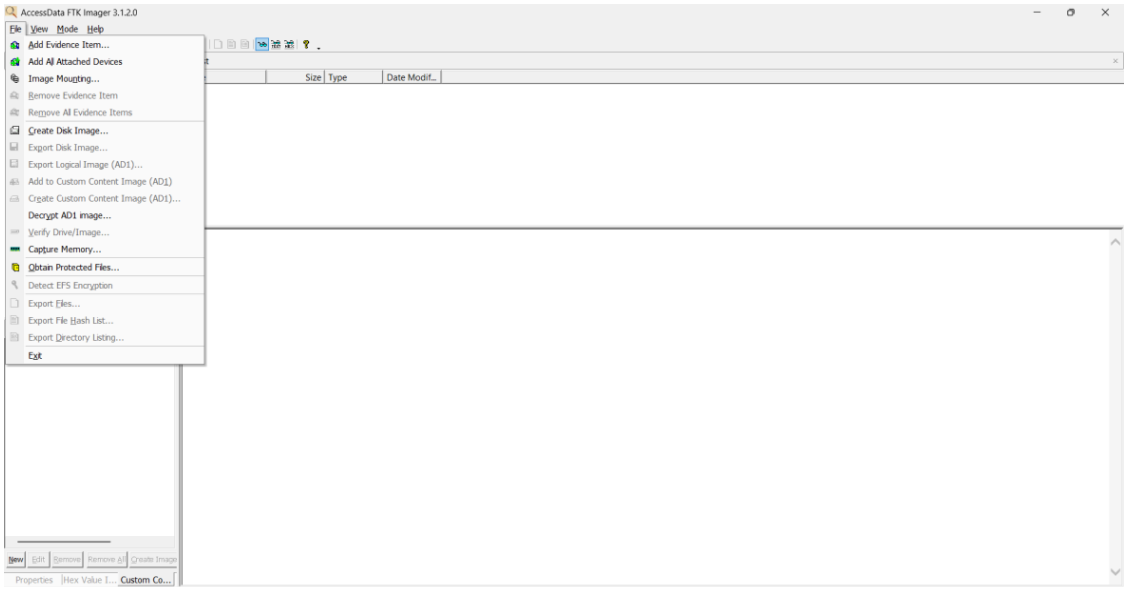


Fig. 5.2 FTK Imager – File → Create Disk Image
Timestamp: 11-10-2026, 10:35:35

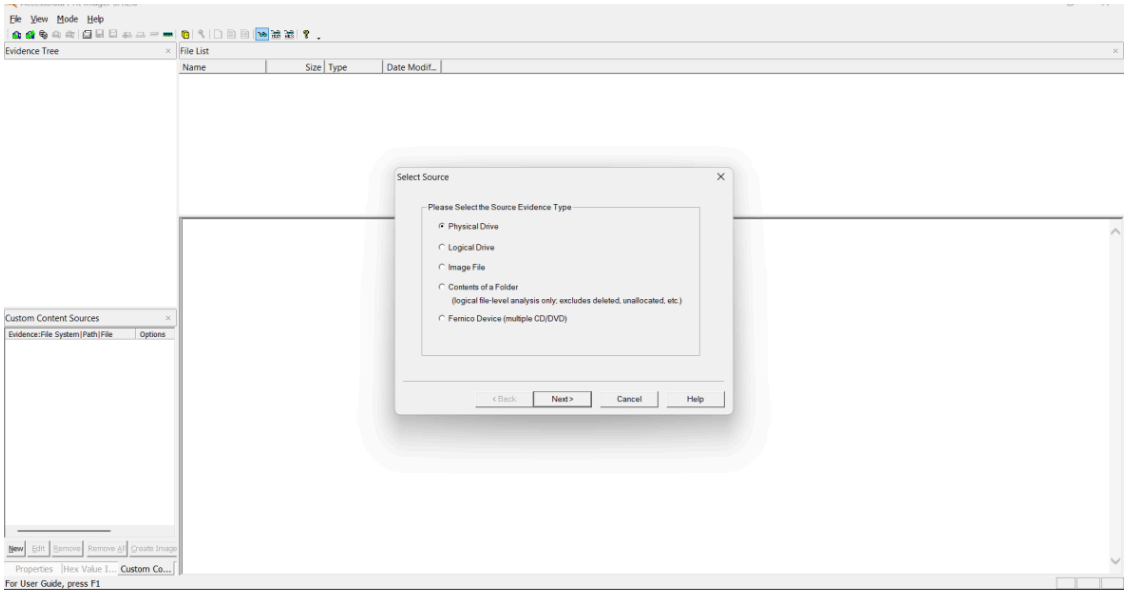


Fig. 5.3 Selecting the source evidence type
Timestamp: 11-10-2026, 10:37:06

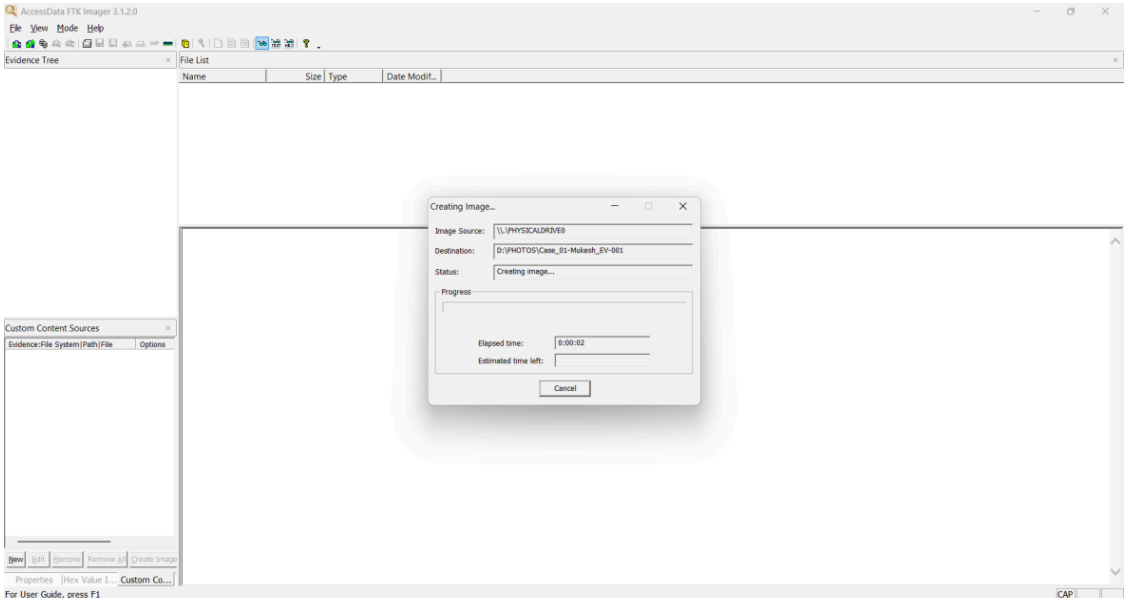


Fig. 5.4 Image creation in progress (first acquisition)
Timestamp: 11-10-2026, 10:40:25

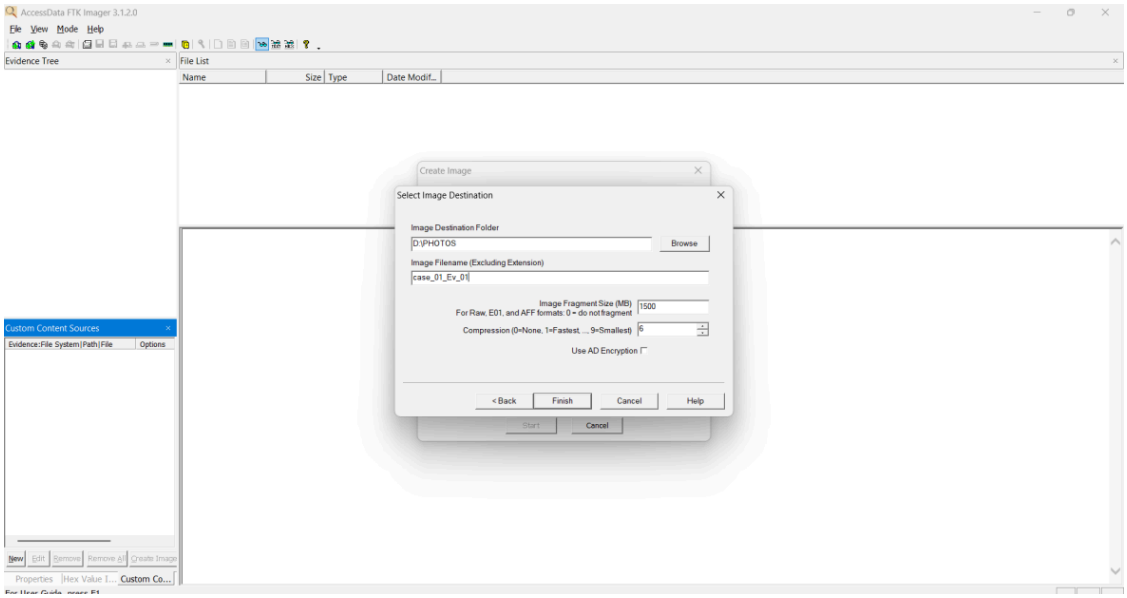


Fig. 5.5 Image destination (second acquisition)
Timestamp: 11-10-2026, 10:45:16

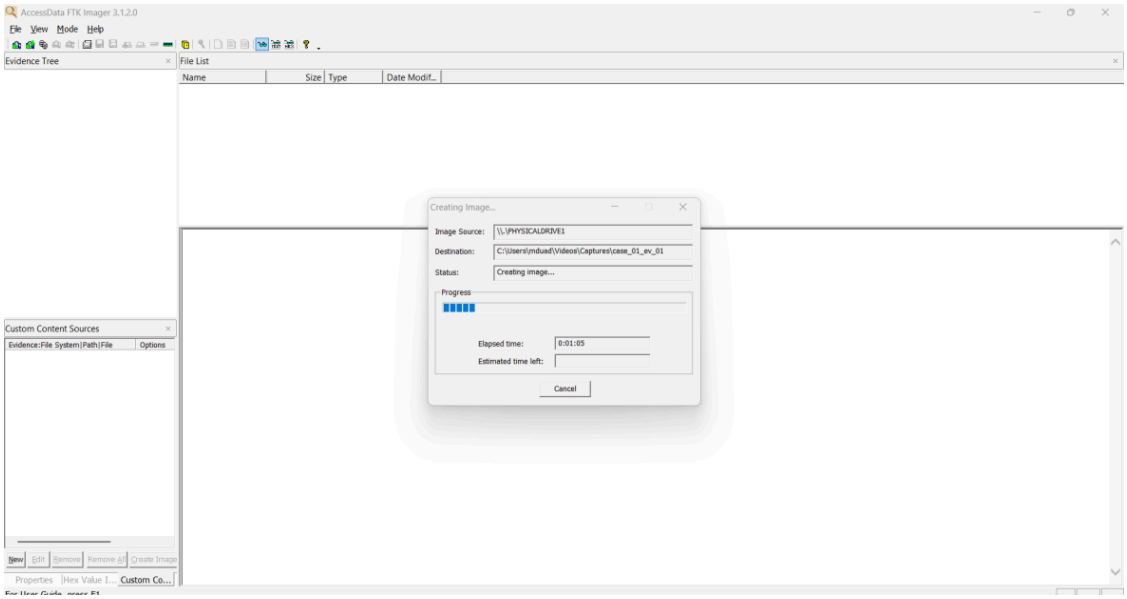


Fig. 5.6 Image creation in progress (second acquisition)
Timestamp: 11-10-2026, 10:47:49

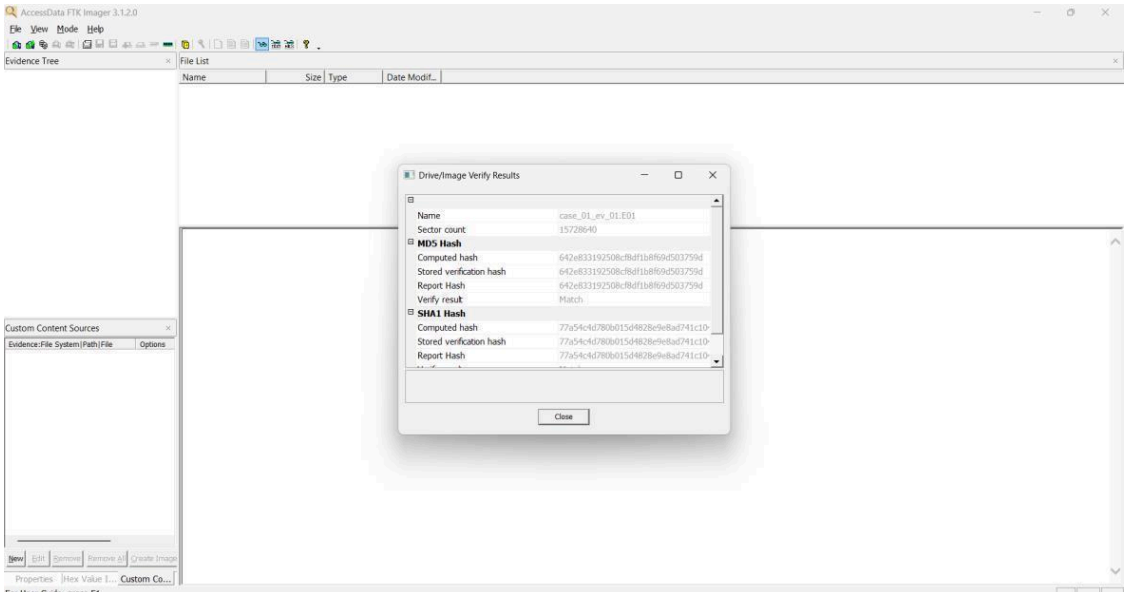


Fig. 5.7 Drive/Image Verify Results – sector count and hash match
Timestamp: 11-10-2026, 10:50:12

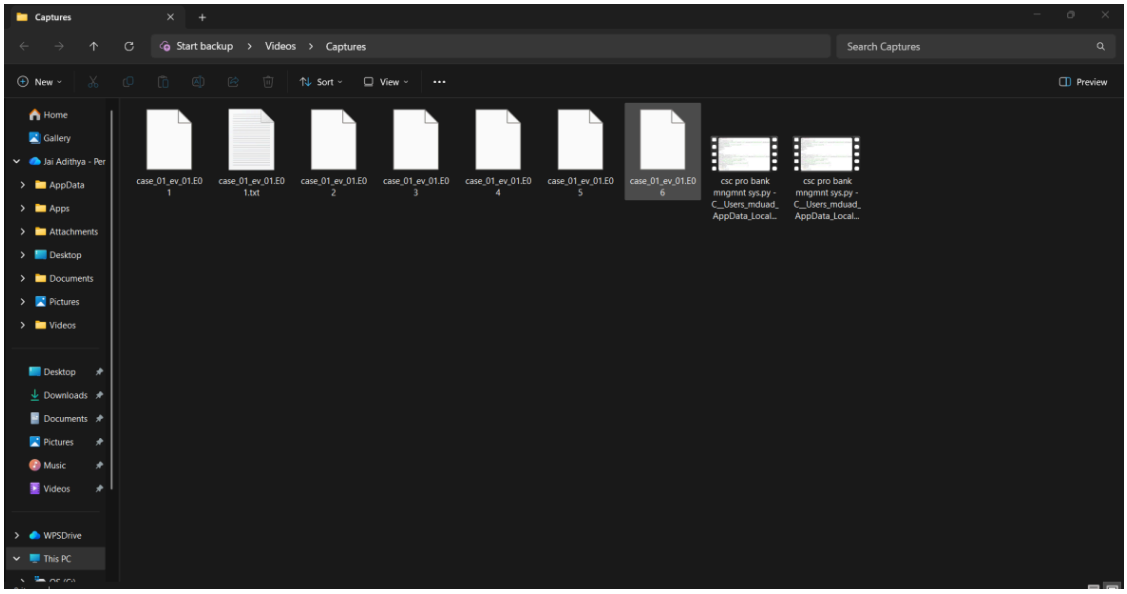


Fig. 5.8 Generated E01 image segments
Timestamp: 11-10-2026, 10:52:41

Result

Thus, a bit-by-bit image of the drive was acquired with FTK Imager as segmented E01 files. The MD5 and SHA-1 verification returned a match with no bad sectors, confirming that the image is an exact copy of the source and is suitable for examining deleted data.

EXPERIMENT – 6

Find Last Connected USB on the System (USB Forensics)

Aim

To identify the last connected USB device on a Windows system using USBDeview.

Procedure

- Downloaded USBDeview (NirSoft) and ran it.
- Reviewed the list of connected and previously connected USB devices (Fig. 6.1).
- Connected the USB flash drive and located its entry, 'Mass Storage – Generic Flash Disk USB Device'.
- Opened its properties to record the serial number, vendor and product IDs, drive letter and connection time (Fig. 6.2).
- Recorded the details as evidence (Table 6.1).

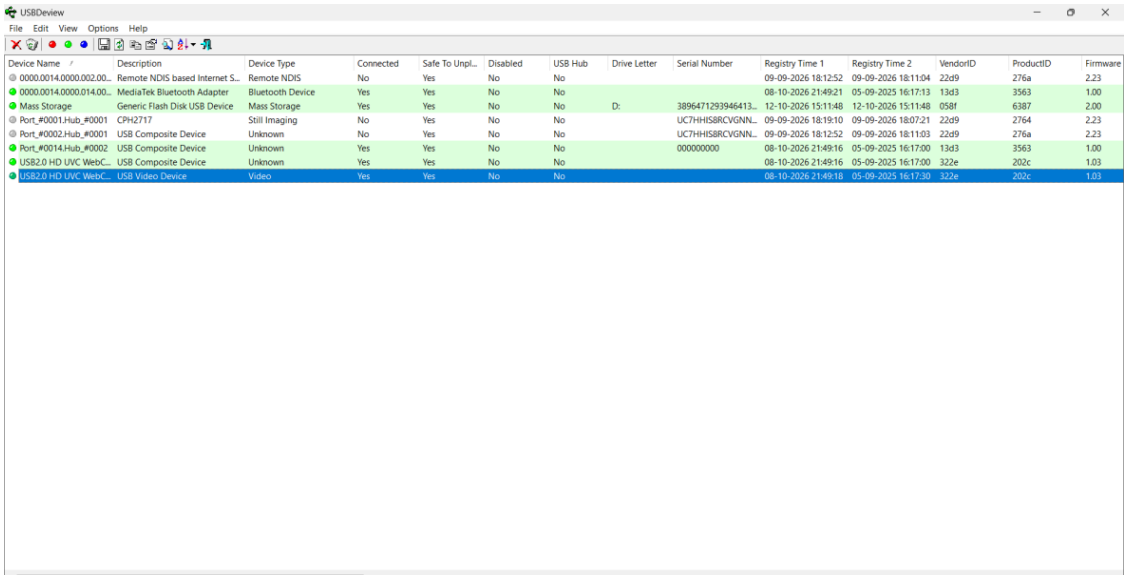


Fig. 6.1 USBDeview – connected and previously used USB devices
Timestamp: 11-10-2026, 12:29:06

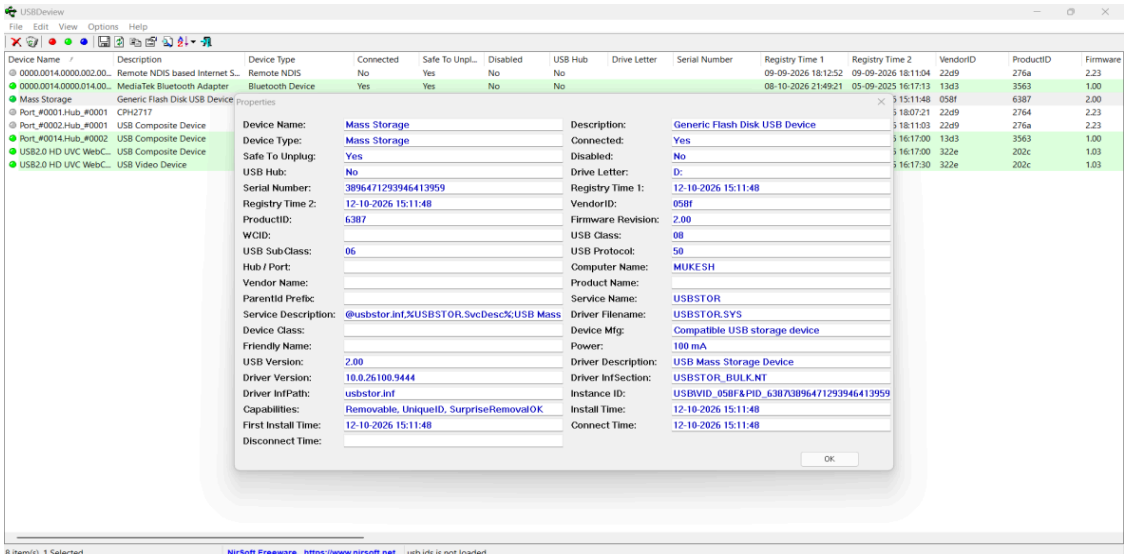


Fig. 6.2 USBDeview – properties of the mass-storage device
Timestamp: 11-10-2026, 12:34:04

Table 6.1 Details of the last connected USB device

Field	Value
Device name / type	Mass Storage
Description	Generic Flash Disk USB Device
Serial number	243567129836726212
Vendor ID / Product ID	023f / 7648
Drive letter	D:
Connected (Registry Time 1 and 2)	11-10-2026 12:23:48
Computer name	MUKESH
Driver	USBSTOR.SYS (USB Mass Storage Device)

Result

Thus, the last connected USB mass-storage device was identified as a Generic Flash Disk (VID 023f, PID 7648, serial 243567129836726212), mounted as drive D: and connected on 11-10-2026 at 12:23:48.

EXPERIMENT – 7

View Last Activity of Your PC

Aim

To view recent user activity and system events using LastActivityView.

Procedure

- Downloaded and ran LastActivityView (NirSoft).
- Observed the columns – Action Time, Description, Filename, Full Path, More Information, File Extension and Data Source.
- Reviewed recent entries such as executed programs (chrome.exe, usbdeview.exe), opened files and folders, and system tasks (Fig. 7.1).
- Opened Properties of a record to see the exact action time and data source such as a Prefetch file (Fig. 7.2).
- Used the timeline to reconstruct the sequence of activity.

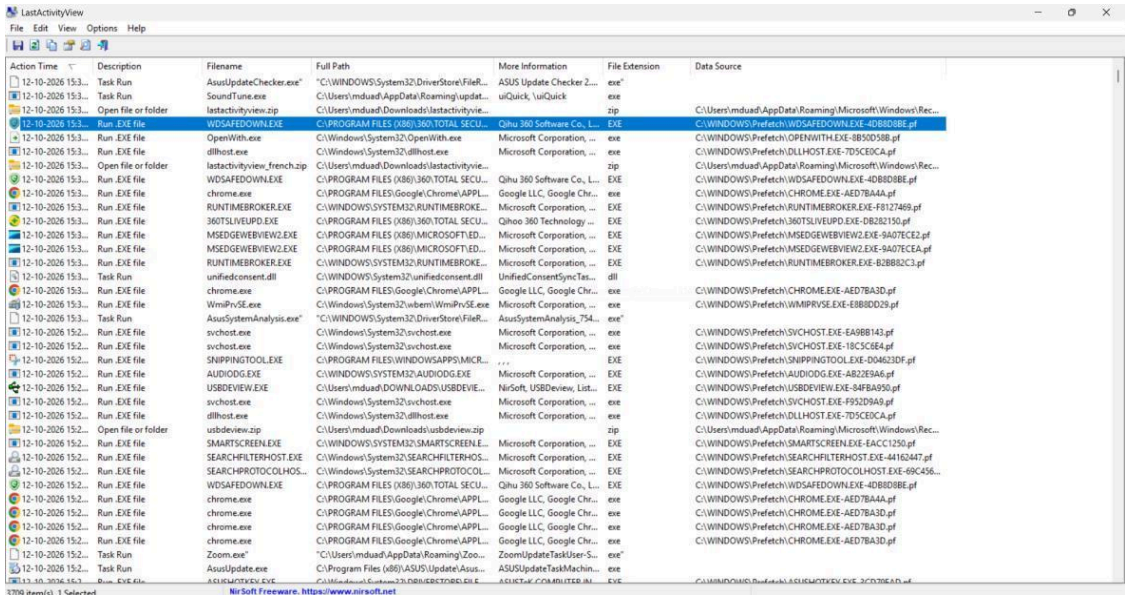


Fig. 7.1 LastActivityView – recent user actions and system events
Timestamp: 11-10-2026, 01:00:52

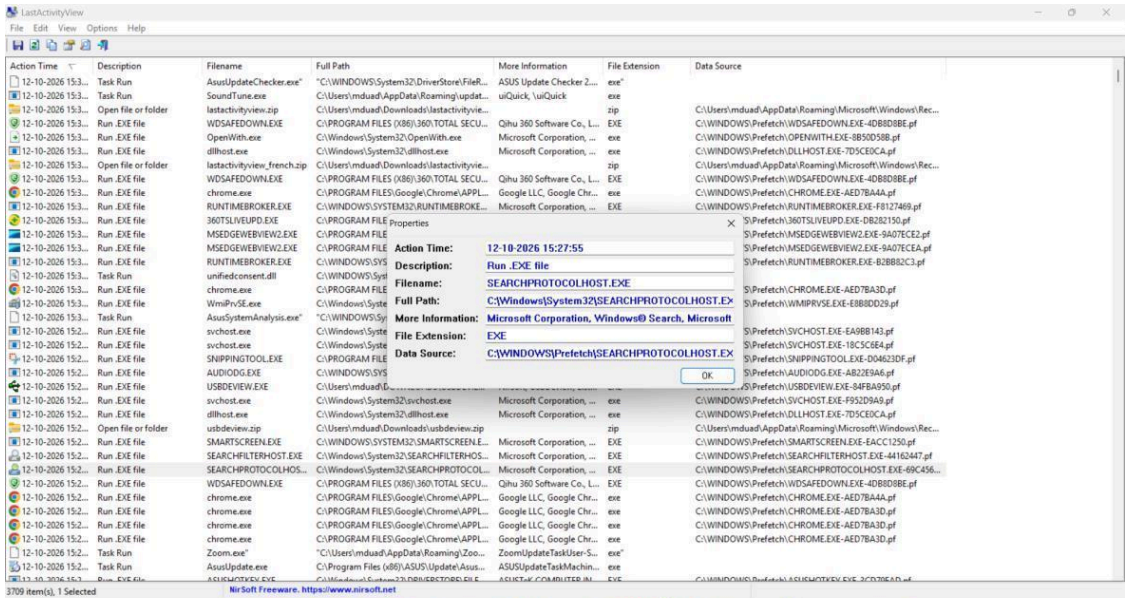


Fig. 7.2 Properties of a record – SEARCHPROTOCOLHOST.EXE executed at 12-10-2026 15:27:55, source: Prefetch
Timestamp: 11-10-2026, 01:04:23

Result

Thus, recent activity on the system was viewed with LastActivityView. The tool listed executed programs, opened files and system events with timestamps and data sources, giving a timeline for forensic analysis.

EXPERIMENT – 8

Extraction of Browser Artifacts

Aim

To extract browser artifacts such as browsing history from a system using browser forensic tools.

Procedure

- Ran ChromeHistoryView, which reads Chrome's history file and lists visited pages with URL, title, visit time, visit count, typed count, referrer and visit ID (Fig. 8.1).
- Used Find to search the history for the keyword 'selfmade' (Fig. 8.2).
- Opened the properties of an entry to view the visit time, visit ID, profile, transition type and the history file path (Fig. 8.3).
- ChromeCacheView, IEHistoryView and IECacheView can be used the same way to read cache and Internet Explorer data.

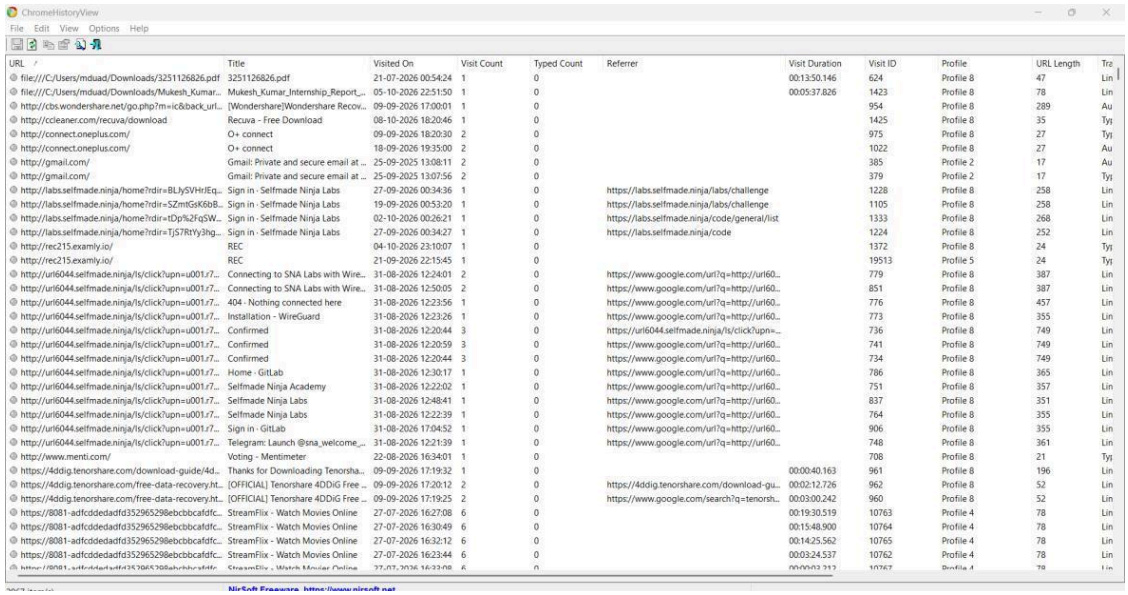


Fig. 8.1 ChromeHistoryView – history of visited pages (1,745 items)
Timestamp: 11-10-2026, 02:03:08

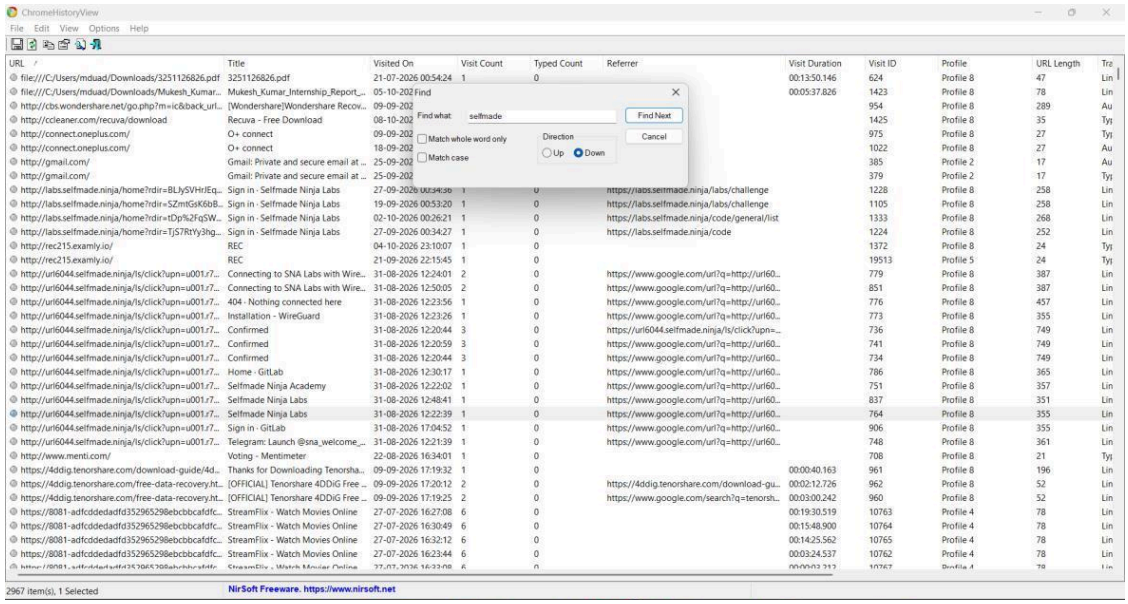


Fig. 8.2 Searching the history for a keyword
Timestamp: 11-10-2026, 02:06:33

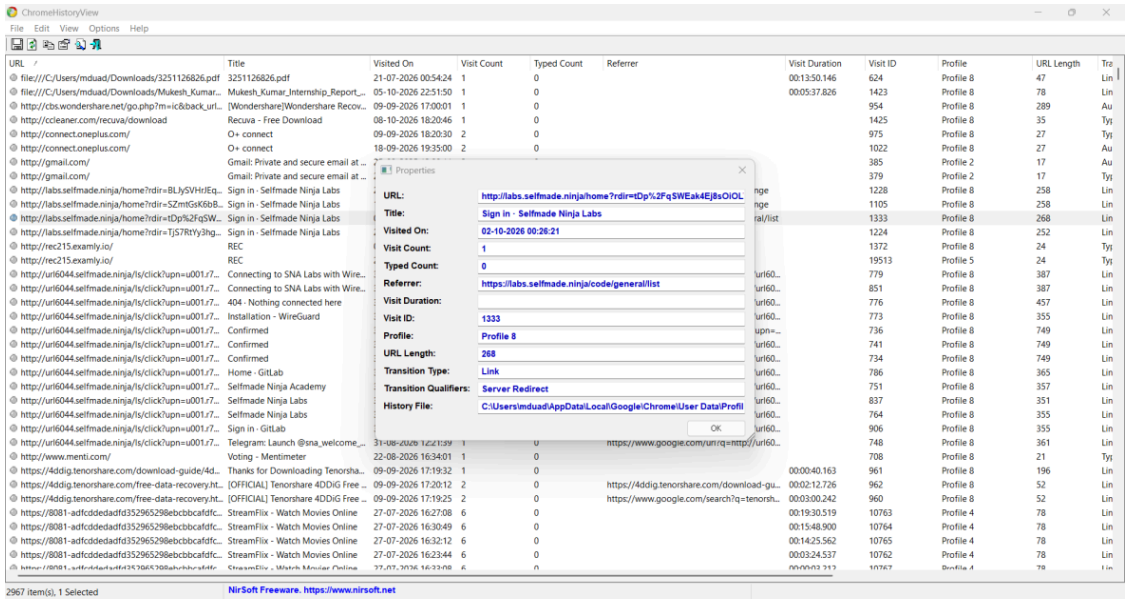


Fig. 8.3 Details of a selected entry – visited 13-09-2026 01:00:46, Visit ID 2434, Profile 3
Timestamp: 11-10-2026, 02:08:46

Result

Thus, browser history was extracted with ChromeHistoryView. The visited URLs, visit times, visit counts and referrers were obtained and can serve as evidence of user activity.

EXPERIMENT – 9

Comparison of Two Files by Compare It Software

Aim

To compare two files with Compare It and generate a report of the differences.

Procedure

- Installed and opened Compare It.
- Selected the first file and then the second file (Fig. 9.1).
- Reviewed the side-by-side view, where differing sections are highlighted (Fig. 9.2).
- Generated a printable comparison report and saved it as a PDF (Fig. 9.3).

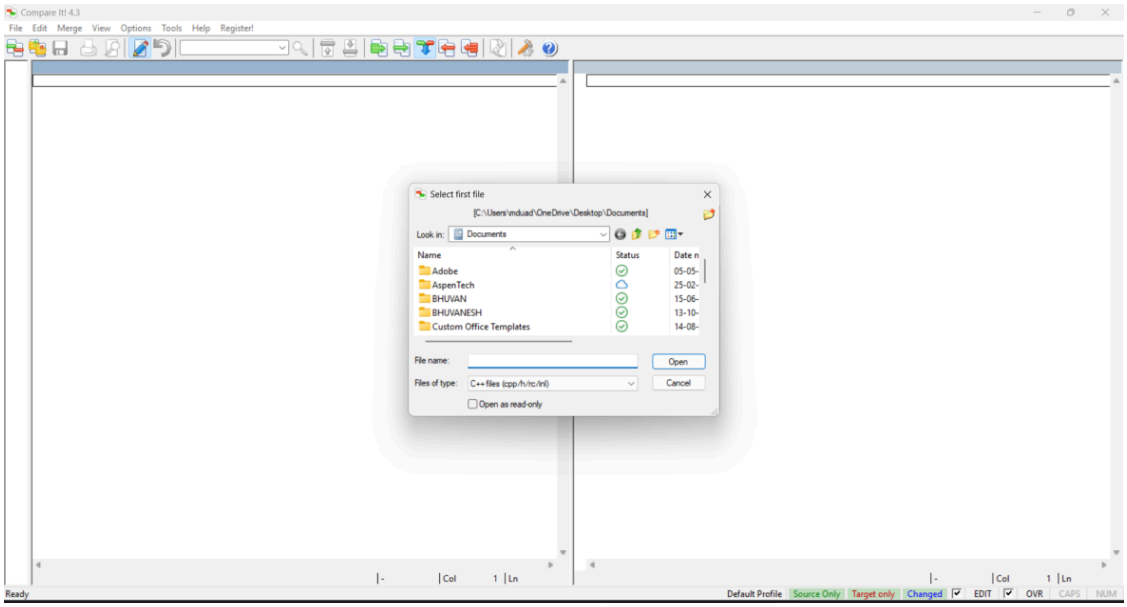


Fig. 9.1 Compare It – selecting the first file
Timestamp: 11-10-2026, 06:04:35

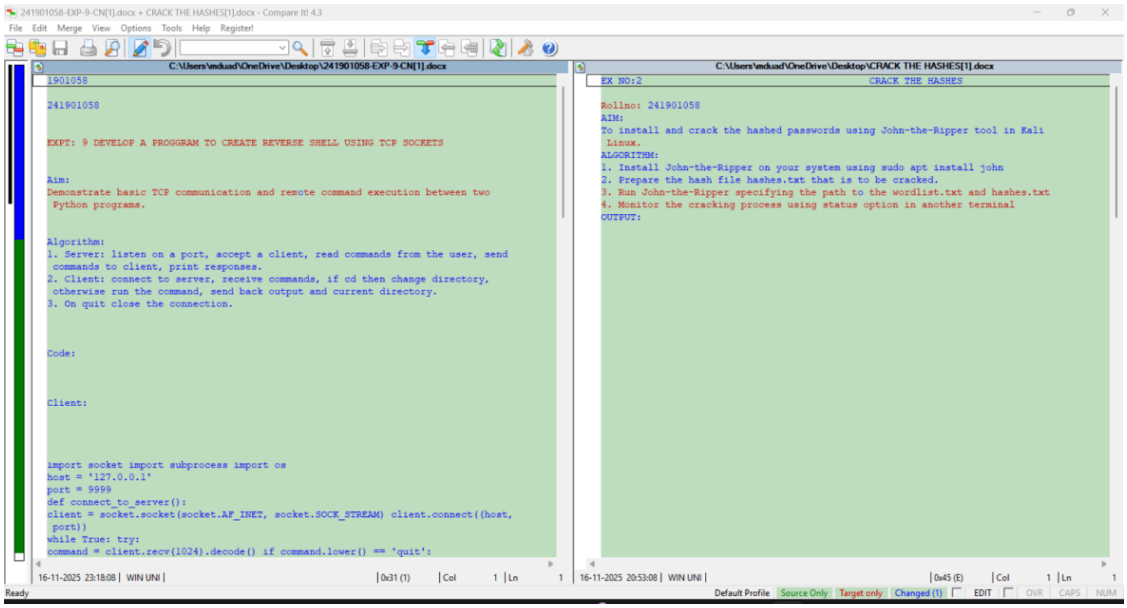


Fig. 9.2 Side-by-side comparison of the two files
Timestamp: 11-10-2026, 06:08:17

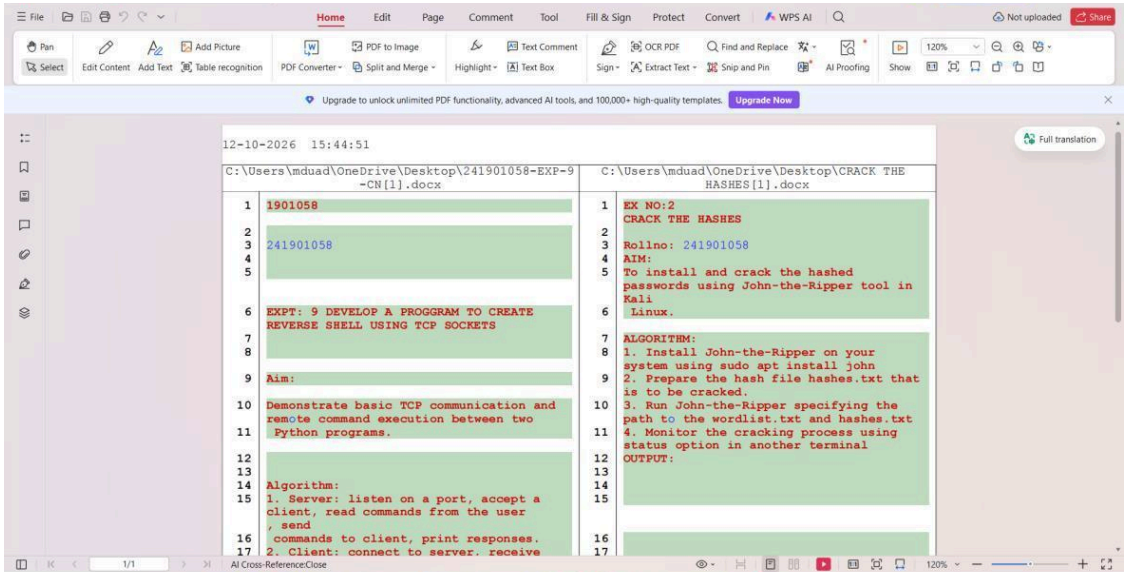


Fig. 9.3 Generated comparison report (cdfreport.pdf)
Timestamp: 11-10-2026, 06:15:56

Result

Thus, two files were compared with Compare It. The differences were highlighted and a comparison report was generated.

EXPERIMENT – 10

Collecting Email Evidence in Victim PC

Aim

To acquire the physical memory of a Windows system and extract e-mail artifacts from it.

Procedure

- **Memory acquisition:** Ran DumpIt, which dumps physical memory on 32-bit and 64-bit Windows and can be run from a USB drive. Confirmed the destination path and typed ‘y’ to proceed (Figs. 10.1 and 10.2).
- **Feature extraction:** Opened Bulk Extractor, selected the memory image as input and an output folder, and started the scan.
- **Review:** Opened the generated feature files – email.txt, email_histogram.txt, url.txt and telephone.txt – in Bulk Extractor Viewer (Figs. 10.3 and 10.4).

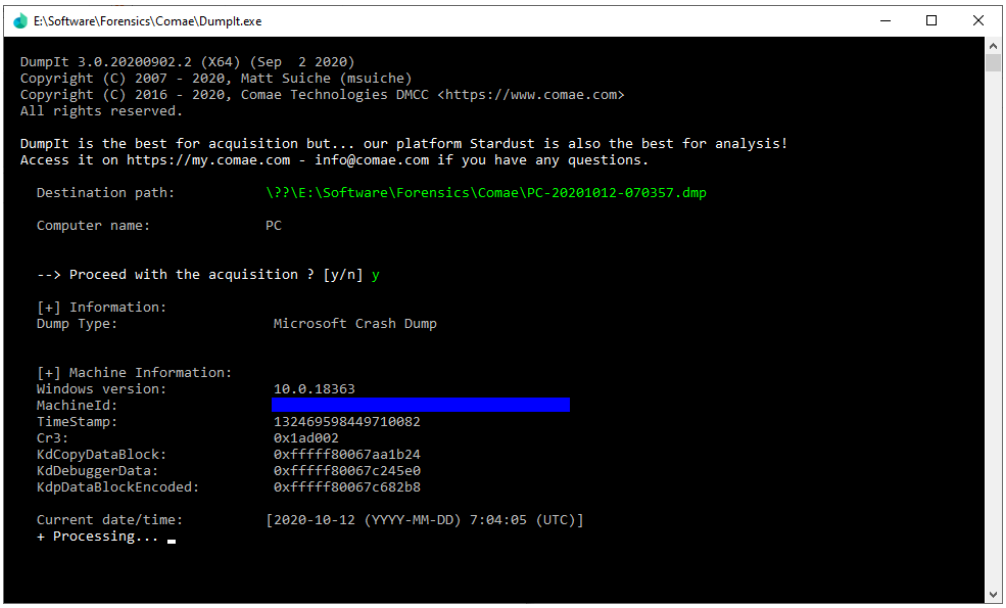


Fig. 10.1 DumpIt – acquisition in progress (reference illustration)

Time Stamp: 12-10-2026, 07:04:49

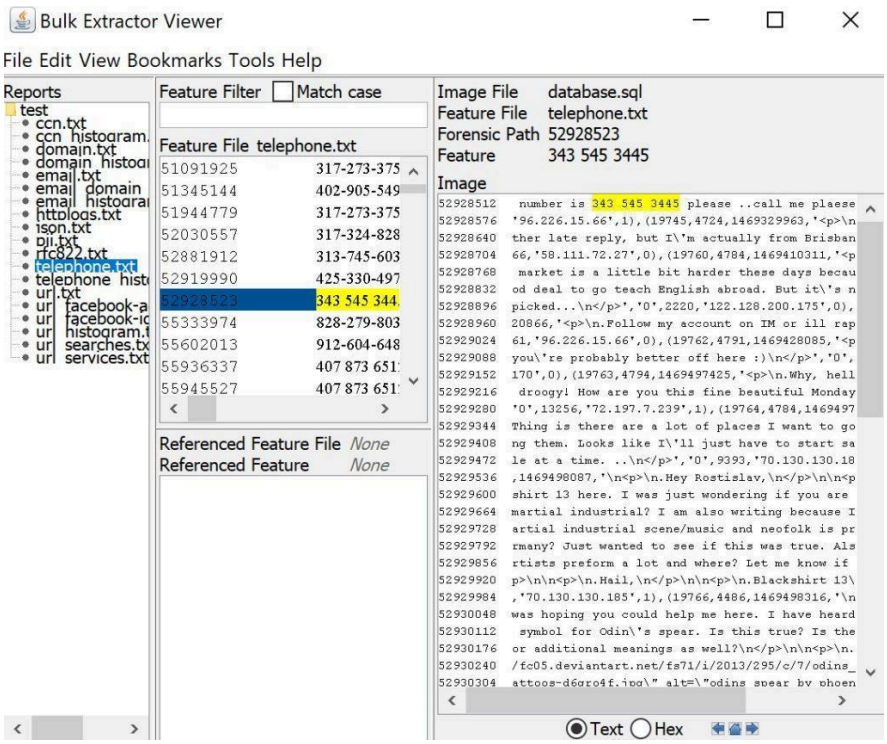


Fig. 10.2 Bulk Extractor Viewer – feature files and extracted entries (reference illustration)

Time Stamp: 12-10-2026, 07:07:23

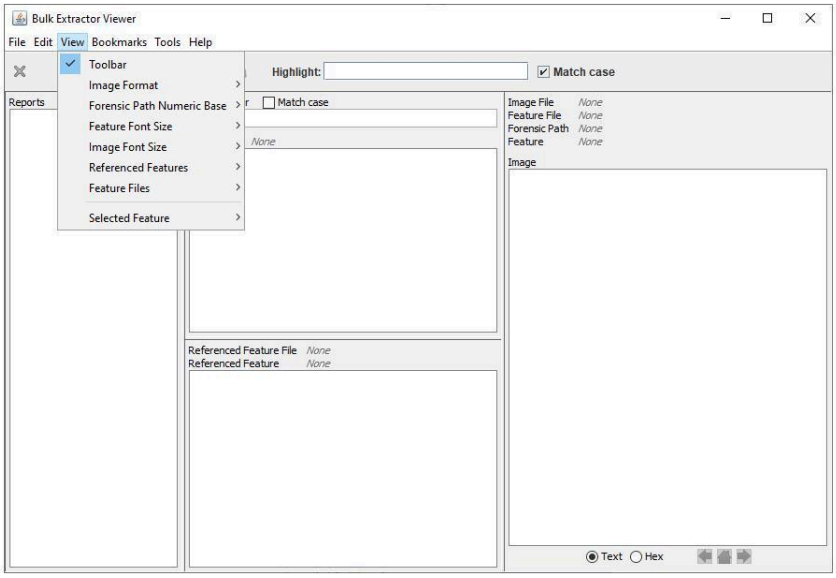


Fig. 10.3 Bulk Extractor Viewer – interface
Timestamp: 12-10-2026, 07:11:55

Result

Thus, the physical memory of the system was acquired with DumpIt and processed with Bulk Extractor, which produces feature files such as email.txt, email_histogram.txt, url.txt and telephone.txt for examining e-mail addresses and related artifacts.

EXPERIMENT – 11

Hiding and Extracting a Text File behind an Image / Audio File using Command Prompt

Aim

To hide a text file, a RAR archive and a ZIP archive behind an image or audio file using the Command Prompt, and to recover the hidden data.

Procedure

- **Prepare:** Placed the file to hide (sample.txt) and the carrier image (sample1.jpg) in one folder .
- **Merge:** Opened Command Prompt, changed to the folder and ran the binary copy: copy /b sample1.jpg + sample.txt result.jpg .
- **Check:** The output file result.jpg is created and still opens as an image .
- **Extract:** Opened result.jpg in Notepad; the hidden text appears after the image data .
- **RAR in image:** copy /b sample1.jpg + sample.rar result.jpg – the archive opens in WinRAR.
- **ZIP in audio:** copy /b sound.mp3 + new.zip newfile.mp3 – the data is recovered by renaming to .zip or opening with an archive tool.
- **Linux:** cat sound.mp3 secret.zip > newfile.mp3

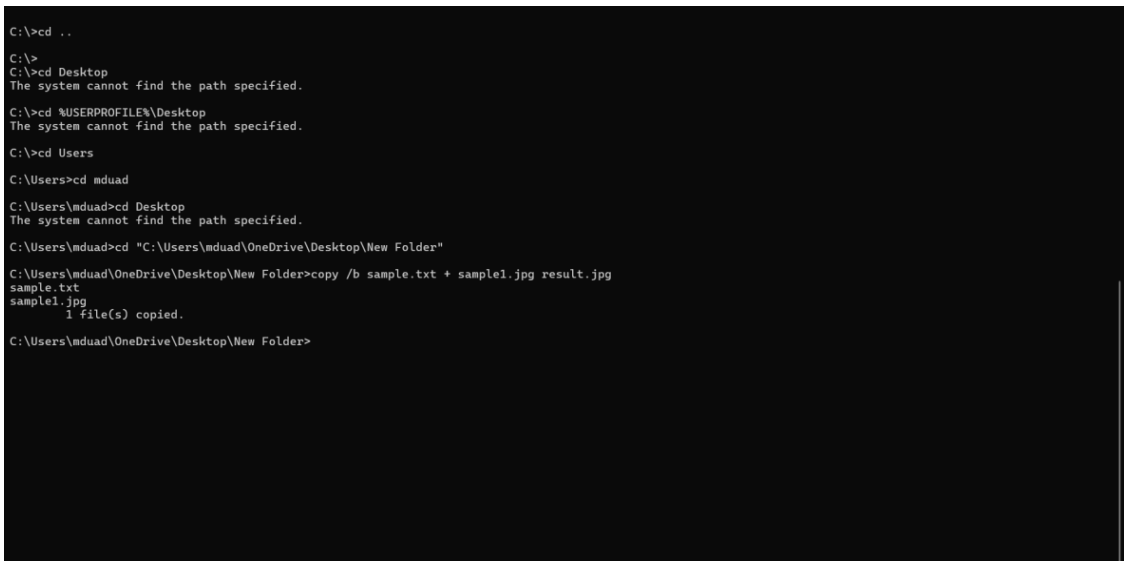


Fig. 11.1 Command Prompt – merging files with copy /b
Timestamp: 12-10-2026, 08:37:19



Fig. 11.2 result.jpg opened in Notepad
Timestamp: 12-10-2026, 08:39:23

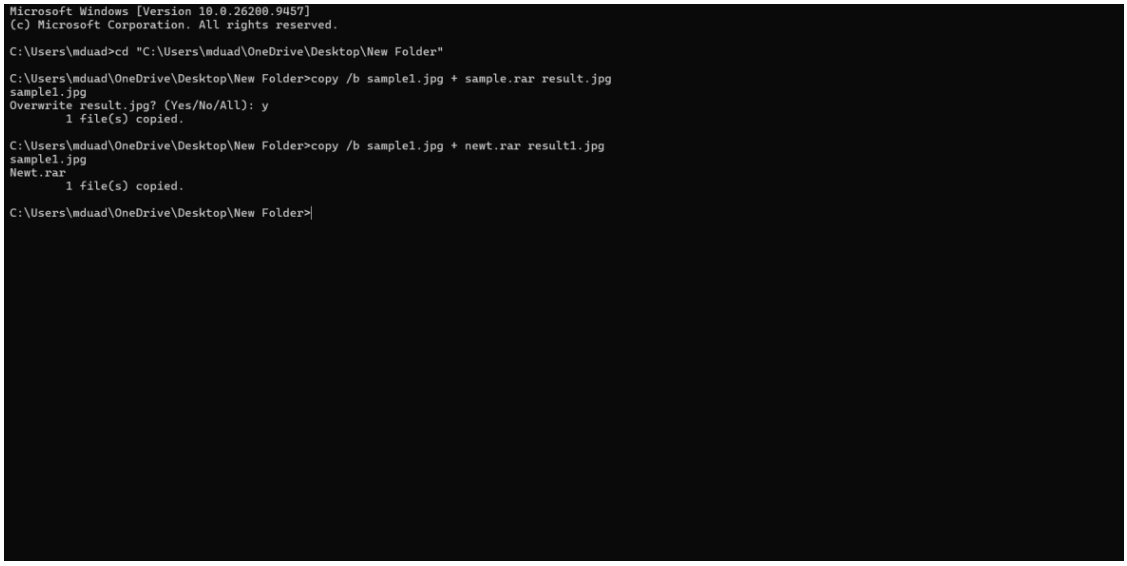


Fig. 11.3 Hiding RAR archives behind the image
Timestamp: 12-10-2026, 08:44:24

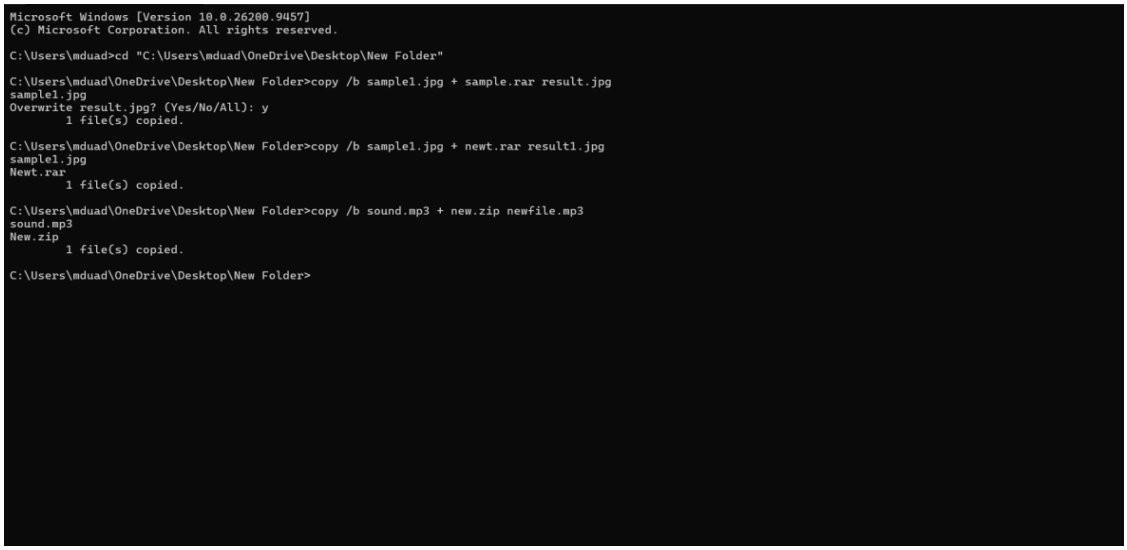


Fig. 11.4 Merging the ZIP archive with the audio file
Timestamp: 12-10-2026, 08:49:31

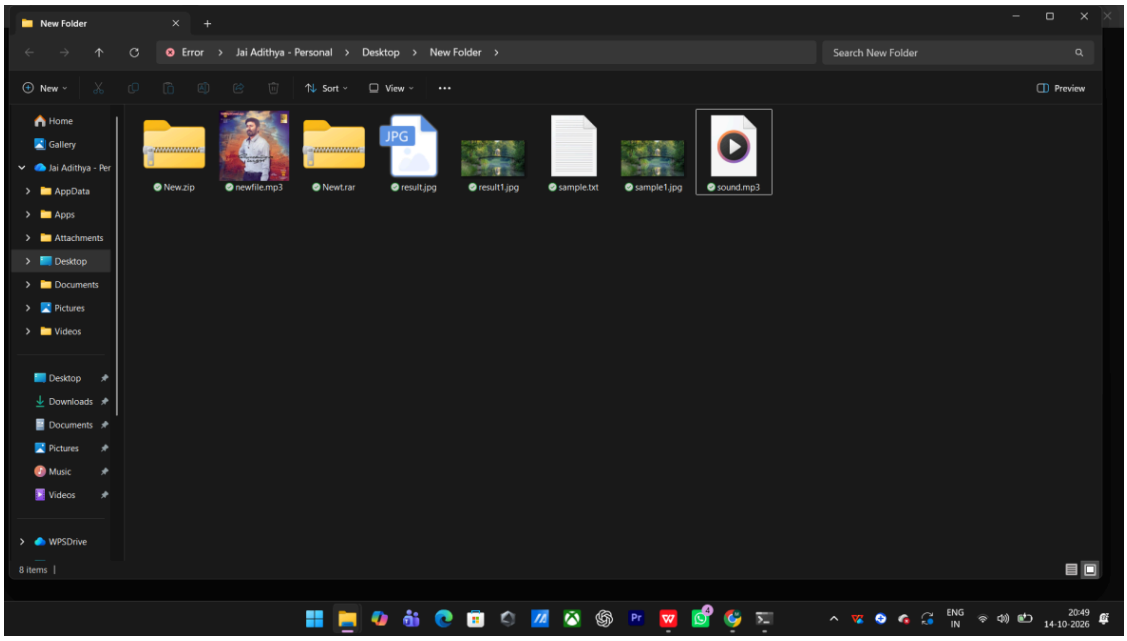


Fig. 11.5 Output audio file newfile.mp3
Timestamp: 12-10-2026, 08:49:43

Result

Thus, a text file, a RAR archive and a ZIP archive were appended to an image and an audio file using the Command Prompt. The output files remain playable or viewable as normal media, which shows why appended data must be considered during forensic examination of media files.

EXPERIMENT – 12

Extraction of EXIF Data from Image Files using ExifReader

Aim

To extract EXIF (Exchangeable Image File Format) metadata from an image file using ExifReader.

Introduction

Seized devices often contain images. Most digital cameras and phones embed EXIF metadata in them, recording the camera make and model, date and time, exposure settings, flash use and sometimes the GPS location. This helps an investigator establish who, what, when and where for an image. ExifReader is a simple Windows tool for viewing this data.

Procedure

- Ran ExifRead.exe and clicked ‘Open’ to select a JPEG image.
- The thumbnail appears on the left and the EXIF data on the right (Fig. 12.1).
- Noted the relevant fields: Make, Model, DateTime, DateTimeOriginal, ExposureTime, FNumber, ISO speed, FocalLength, Flash and GPS data where present.

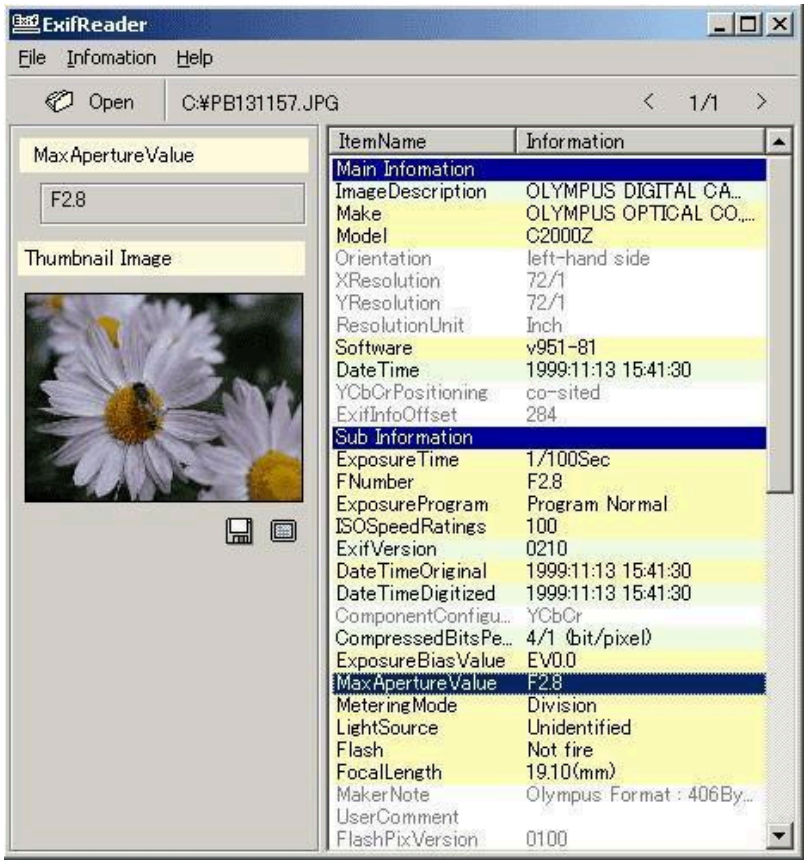


Fig. 12.1 ExifReader – EXIF data of a sample JPEG (reference illustration)

Result

Thus, EXIF data was extracted with ExifReader. The camera make and model, capture date and time and exposure settings were obtained, which can link an image to its source device.